

SmolVLM for Dense Video Captioning

Week 11: Internship Update

Parameters and VideoMME

- 694 unique videos

Qwen2.5-VL-3B-Instruct

- Video categories: center and first

Video Type	first	center	fps:fps:min:max	maxinfo:input:max	csta:input:max
Short	38.5%	43.6%	%	%	%
Medium	48.2%	50.4%	%	%	%
Long	43.8%	46.8%	%	%	%
Entire dataset	44.9%	47.8%	%	%	%

Table 9: VideoMME results over Video Category: Knowledge, Model: Qwen2.5

Video Type	first	center	fps:fps:min:max	maxinfo:input:max	csta:input:max
Short	47.4%	48.2%	%	%	%
Medium	41.2%	27.5%	%	%	%
Long	39.1%	42.0%	%	%	%
Entire dataset	43.6%	41.9%	%	%	%

Table 10: VideoMME results over Video Category: Film & Television, Model: Qwen2.5

Video Type	first	center	fps:fps:min:max	maxinfo:input:max	csta:input:max
Short	49.0%	46.3%	%	%	%
Medium	29.9%	36.7%	%	%	%
Long	43.9%	43.0%	%	%	%
Entire dataset	40.7%	41.9%	%	%	%

Table 11: VideoMME results over Video Category: Sports Competition, Model: Qwen2.5

Qwen2.5-VL-3B-Instruct

- Video categories: center and first

Video Type	first	center	fps:fps:min:max	maxinfo:input:max	csta:input:max
Short	46.2%	57.3%	%	%	%
Medium	43.3%	39.2%	%	%	%
Long	34.2%	43.9%	%	%	%
Entire dataset	41.3%	46.7%	%	%	%

Table 12: VideoMME results over Video Category: Artistic Performance, Model: Qwen2.5

Video Type	first	center	fps:fps:min:max	maxinfo:input:max	csta:input:max
Short	45.1%	46.6%	%	%	%
Medium	38.6%	36.2%	%	%	%
Long	37.2%	40.6%	%	%	%
Entire dataset	40.3%	41.1%	%	%	%

Table 13: VideoMME results over Video Category: Life Record, Model: Qwen2.5

Video Type	first	center	fps:fps:min:max	maxinfo:input:max	csta:input:max
Short	29.6%	29.6%	%	%	%
Medium	50.0%	46.7%	%	%	%
Long	36.7%	36.7%	%	%	%
Entire dataset	39.1%	37.9%	%	%	%

Table 14: VideoMME results over Video Category: Multilingual, Model: Qwen2.5

Qwen2.5-VL-3B-Instruct

- Video categories: center and first

Video Type	first	center	fps:fps:min:max	maxinfo:input:max	csta:input:max
Short	45.5%	47.8%	%	%	%
Medium	40.2%	39.5%	%	%	%
Long	39.7%	43.1%	%	%	%
Entire dataset	41.7%	43.4%	%	%	%

Table 15: VideoMME results over frame sampling techniques: Model: Qwen2.5

Video Type	first	center	fps:fps:min:max	maxinfo:input:max	csta:input:max
Temporal Perception	39.0%	29.3%	%	%	%
Spatial Perception	48.8%	48.8%	%	%	%
Attribute Perception	47.1%	47.8%	%	%	%
Action Recognition	40.4%	42.6%	%	%	%
Object Recognition	42.2%	41.8%	%	%	%
OCR Problems	39.8%	42.6%	%	%	%
Counting Problem	29.6%	26.8%	%	%	%
Temporal Reasoning	34.0%	33.3%	%	%	%
Spatial Reasoning	48.7%	64.1%	%	%	%
Action Reasoning	40.8%	42.2%	%	%	%
Object Reasoning	43.8%	47.0%	%	%	%
Information Synopsis	51.8%	59.4%	%	%	%

Table 16: VideoMME results over task categories: Model: Qwen2.5